

In the next section, we will create charts using Python.

Charts using Pandas

In this section, we will create charts using Python script. Let's create chart data with pivot, as follows (see Table 6):

```
1. chartData = pd.pivot_table(resultData, index=config.
get('Pivot', 'index-level-0').split(',').columns=config.
get('Pivot', 'columns').split(',').values=config.get('Pivot',
'values'), aggfunc=len, margins=True, fill_value=0)
```

Project manager	Completed	Pending	All
Jayesh P	2	1	3
Mukund	1	1	2
Radhika M	1	4	5
All	4	6	10

Table 6: Chart data

You can create and plot a chart based on the chart data given in Table 6:

```
1. ax = chartData.plot(kind='bar')
2. for p in ax.patches:
3.     ax.annotate(str(p.get_height()), (p.get_x() * 1.005,
p.get_height() * 1.005))
4. plt.show()
```

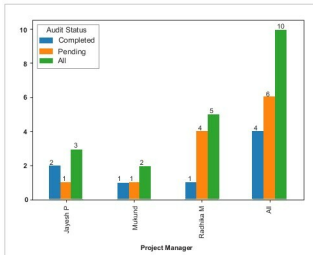


Figure 5: Plot a chart

The following script helps to create a bar chart that is editable in the Excel file:

```
1. # pip install xlswriter
2. writer = pd.ExcelWriter('Results.xlsx',
```

```
engine='xlswriter')
3. #store data frames in a Dictionary object, where the key
is the sheet name
4. frames = {'MasterData': chartData}
5. #store data frames in a Dictionary object, where the key
is the sheet name
6. for sheet, frame in frames.items():
7.     frame.to_excel(writer, sheet_name = sheet)
8.
9. worksheet = writer.sheets['MasterData']
10. workbook = writer.book
11. chart = workbook.add_chart({'type': 'column'})
12. # Some alternative colors for the chart.
13. colors = ['#4DAF4A', '#E41A1C', '#377EB8']
14.
15. chart.add_series({
16.     'name': '=MasterData!$C$1',
17.     'categories': '=MasterData!$B$2:$B$6',
18.     'values': '=MasterData!$C$2:$C$7',
19.     'fill': {'color': colors[0]},
20.     'overlap': -10,
21.     'gap': 400,
22.     'data_labels': {'value': True},
23. })
24.
25. chart.add_series({
26.     'name': '=MasterData!$D$1',
27.     'categories': '=MasterData!$B$2:$B$6',
28.     'values': '=MasterData!$D$2:$D$7',
29.     'fill': {'color': colors[1]},
30.     'gap': 400,
31.     'data_labels': {'value': True},
32. })
33.
34. chart.add_series({
35.     'name': '=MasterData!$E$1',
36.     'categories': '=MasterData!$B$2:$B$6',
37.     'values': '=MasterData!$E$2:$E$7',
38.     'fill': {'color': colors[2]},
39.     'gap': 400,
40.     'data_labels': {'value': True},
41. })
42. chart.set_style(11)
43. # Add a chart title
44. chart.set_title({'name': 'Project Manager-wise Data'})
45.
46. # Add x-axis label
47. chart.set_x_axis({
48.
49.     'name': 'Project Manager',
50.     'major_gridlines': {
51.         'visible': True,
```